

Freight Rate prediction Model report

1. Objective

Train model on data/train_test.csv (48,000 labelled loads, January to October 2025) to predict posted_rate for the 12,000 loads in data/validation.csv (November to December 2025) and to produce a 31- day rate forecast for single fixed lane (Lexington to Fort Wayne, Dry Van, 360 miles, 32,000 lb) across December 2025.

2. Data Quality Issues Found

- a. Negative weight values: 292 (0.6%) of 48,000 rows had negative weight, with magnitudes in the same 0 to 47,500 lb range as the valid rows (rows with positive weights). They were treated as entry errors and corrected with abs() instead of being dropped.
- b. Missing values: weight (300 rows) and market_index (374 rows) had nulls in the train_test.csv. Similar small gaps existed in the validation.csv (165 and 249 rows respectively). Weight was imputed with the equipment type median whereas market_index/ quote signal were imputed using the mean of all other loads recorded on the same date since both signals turned out to be date-level market conditions rather than lane-specific values.
- c. Provided distance did not match the great circle distance i.e. the computed distance for the pickup and delivery latitude -longitude pairs (differs by approximately 170 miles on average and up to 580 miles). The latitude and longitude fields looked noisy relative to the named cities, so the model used the provided distance column directly rather than deriving the distance from the coordinates.
- d. December_chart_inputs.csv had no market_index and quote signal columns unlike every other data file. Both signals turned out to be date-level market conditions so daily averages computed from train_test.csv and validation.csv were used to estimate them for the December fixed lane by date.

3. Feature engineering

- a. Distance, weight (cleaned), equipment, pickup and delivery were used directly. Pickup, delivery and equipment were passed as native categorical features to the gradient boosting model rather than splitting into many separate indicator columns.
- b. Market_index and quote signal were used directly as well, imputed by date where missing.
- c. Date derived: month, day_of_week, day_of_year and is_weekend were included in case they captured seasonality not already reflected in the market_index and quote_signal.

4. Train/Validation split approach

Train_test.csv covered January 1 to October 31, 2025, and the actual forecast was from November to December 2025. Which counted as a genuine forecast into unseen future dates and not mere interpolation within a range already seen. As such, a standard random 80/20 split would place holdout rows in time right next to training rows (same week, same market cycle) which is closer to interpolation than real the real forecasting required and would overstate accuracy relative to the task. To avoid this, a chronological (time-based holdout) 80/20 split was used instead.

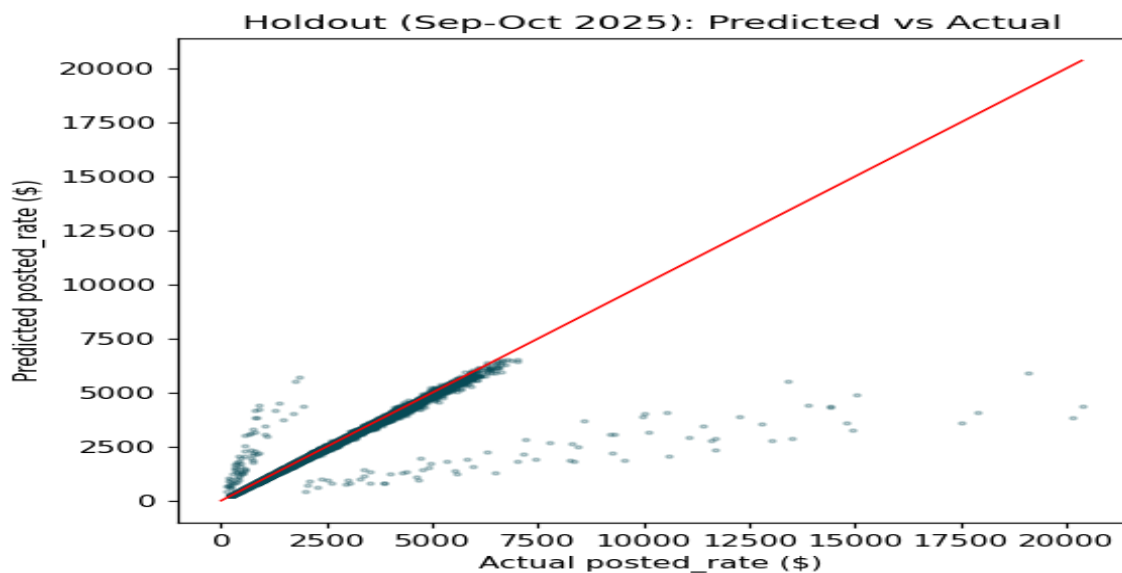
The Chronological split was as follows: train on January 1 to August 31 (80% of the rows), then evaluate on September 1 to October 31 (the last 20% of rows). This mirrors the actual forecast gap where the model is evaluated on dates it has never seen, exactly as it would be when predicting November to December.

5. Model Choice

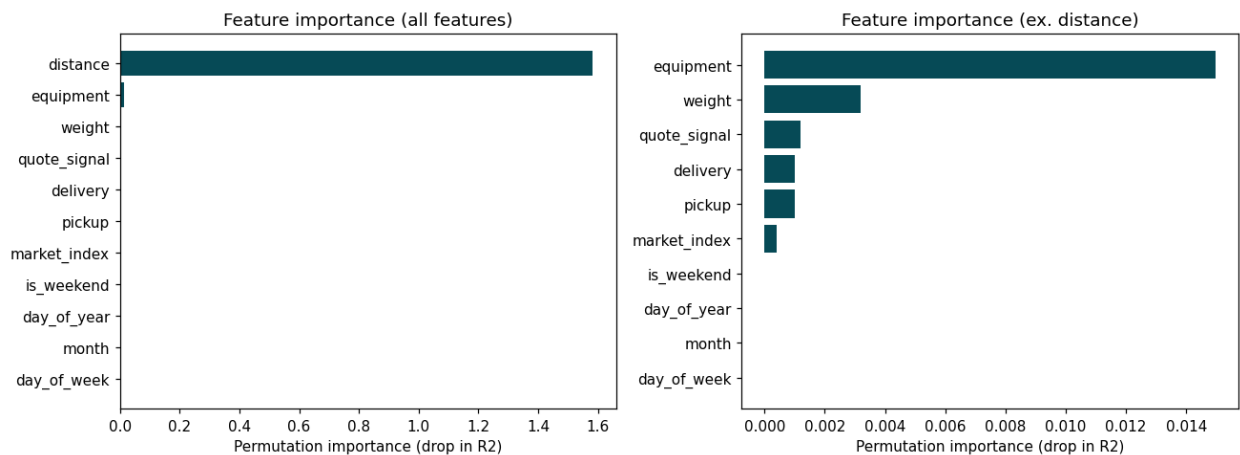
- Model: Scikit -learn HistGradientBoostingRegressor with an absolute-error (MAE) loss.
- Gradient-boosted trees handle the mix of categorical (pickup, delivery, equipment) and numerical (distance, weight, quote signal, market index) features, and the non-linear relationship between them without manual interaction terms.
- MAE loss rather than the default squared error was chosen specifically because of the outlier rows identified.

After model selection, the final model is refit on all 48,000 labelled rows using the iteration count chosen by early stopping during the time-based run, before generating the final predictions, so no labelled data is held back from the model.

6. Results



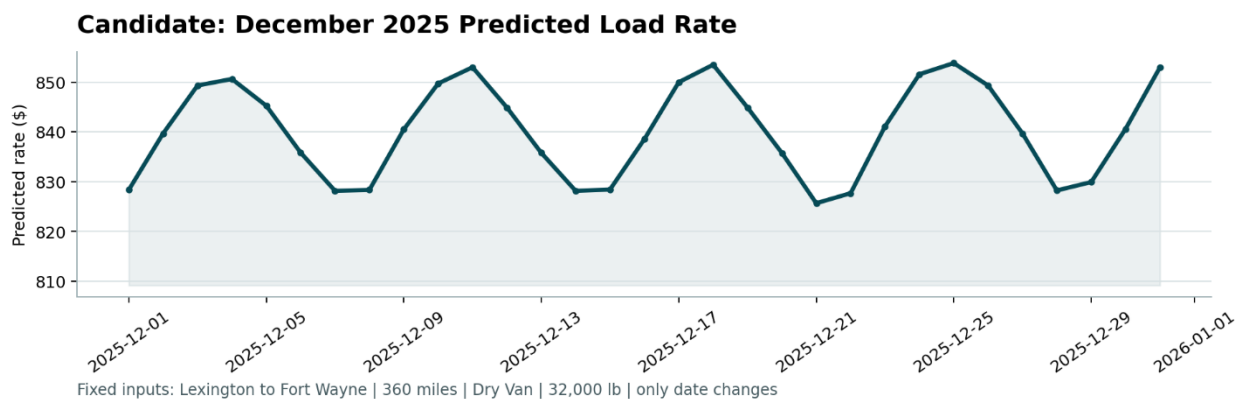
Predicted vs. actual posted_rate on the September to October 2025 holdout. Most points sit tightly on the diagonal and the scattered points below the line are the outlier rows identified.



Distance was by far the most dominant driver of price, followed by equipment type and weight, lane(pickup and delivery), market_index and quote_signal contributed smaller but non-zero adjustments. Calender features (month, day-of-week, day-of-year) which were introduced, had minimal direct importance once market_index and quote_signal which already influenced the weekly cycle, were in the model.

7. December 2025 fixed- lane forecast

For the fixed Lexington to Fort Wayne, Dry van lane, predicted rates for December 2025 ranged from about \$826 to \$854, tracking a repeating weekly pattern (mid-week peak with weekend through) driven by the market_index and quote_signal seasonality described above.



Fixed December predictions chart produced by score.py